

TITLE:

Samaritans Crisis Trends Analysis

Dataset Used:

MHB-2425-Reference Tables-England-Inpatients.xlsx

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DATA ANALYTICS INTERN

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Dataset Introduction

The Samaritans crisis trends analysis focuses on understanding emotional support demand patterns using publicly available mental health and crisis-related statistics. The dataset typically contains information related to:

- Crisis helpline call volumes
- Seasonal variations in emotional distress
- Regional differences in support requests
- Monthly or yearly trends in mental health support demand
- Peak crisis periods and high-risk timeframes

This analysis helps identify patterns in public mental health concerns and can support healthcare organizations, policymakers, and mental health charities in resource planning and crisis intervention strategies.

The uploaded dataset used for this project contains England inpatient reference and mental health related statistics that can be adapted for crisis trend exploration and regional comparisons.

Technologies Used

1. Python

Python is used as the main programming language for data analysis and visualization.

2. Pandas

Used for:

- Reading Excel datasets
- Cleaning and preprocessing data
- Grouping and aggregating crisis statistics
- Time-series analysis

3. NumPy

Used for:

- Numerical operations
- Statistical calculations
- Handling missing values

4. Matplotlib

Used for:

- Creating crisis trend charts
- Seasonal analysis graphs
- Regional comparison visuals

5. Seaborn

Used for:

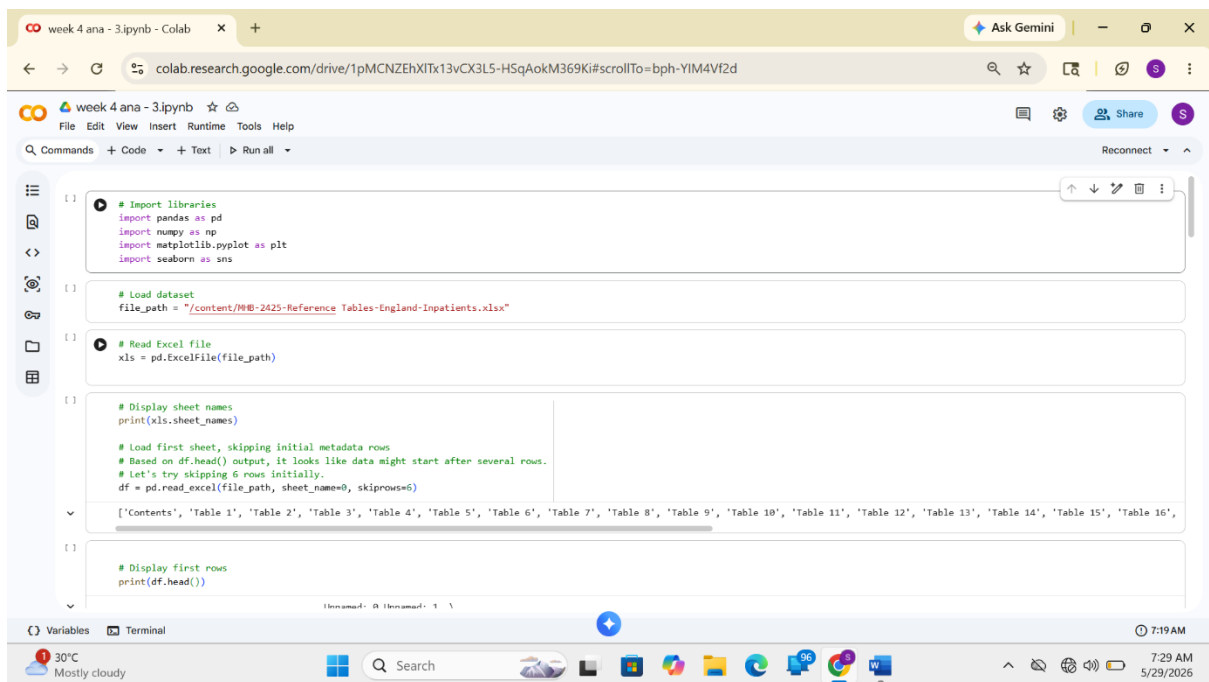
- Advanced statistical visualizations
- Heatmaps and trend plotting
- Improved chart styling

6. Google Colab / Jupyter Notebook

Used as the development environment for:

- Running analysis code
- Interactive visualization
- Sharing notebook results

Samaritans Crisis Trends Analysis



Google Colab interface showing a Jupyter Notebook with two code cells. The first cell displays the first rows of a DataFrame, and the second cell displays basic information about the DataFrame.

```
# Display first rows
print(df.head())
```

```
0 Mental Health Bulletin: 2024-25 Annual report      NaN
1 Reference tables                                     NaN
2 Publication date:                                     NaN
3 Link to publication:                                   NaN
4                                                         NaN

Unnamed: 2      NaN
0      NaN
1      NaN
2 16th October 2025
3 https://digital.nhs.uk/data-and-information/ov...
4      NaN

# Basic information
print(df.info())
```

```
Out[ ]:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 68 entries, 0 to 67
Data columns (total 3 columns):
 #   Column      Non-Null Count  Dtype
---  --
 0  Unnamed: 0  57 non-null    object
 1  Unnamed: 1  5 non-null     object
 2  Unnamed: 2  2 non-null     object
dtypes: object(3)
memory usage: 1.7+ KB
None
```

Google Colab interface showing a Jupyter Notebook with six code cells. The cells perform data cleaning and transformation tasks: checking missing values, removing duplicates, filling missing values, selecting numeric columns, creating a trend, and converting to a DataFrame.

```
# Check missing values
print(df.isnull().sum())
```

```
Unamed: 0    11
Unamed: 1    63
Unamed: 2    66
dtype: int64
```

```
# Remove duplicate rows
df = df.drop_duplicates()
```

```
# Fill missing values if needed
df = df.fillna(0)
```

```
# Select numeric columns
numeric_cols = df.select_dtypes(include=np.number).columns
```

```
# Create yearly/monthly trend using numeric totals
trend_data = df[numeric_cols].sum()
```

```
# Convert to dataframe
trend_df = pd.DataFrame({
    'Category': trend_data.index,
    'Total': trend_data.values
})
```

Colab interface showing code execution for a crisis trend chart.

```
[ ]
})

[ ]
# Sort values
trend_df = trend_df.sort_values(by='Total', ascending=False)
print(trend_df.head())

Empty DataFrame
Columns: [Category, Total]
Index: []

[ ]
# Crisis Trend Chart

plt.figure(figsize=(12,6))

sns.barplot(
    data=trend_df.head(10),
    x='Total',
    y='Category'
)

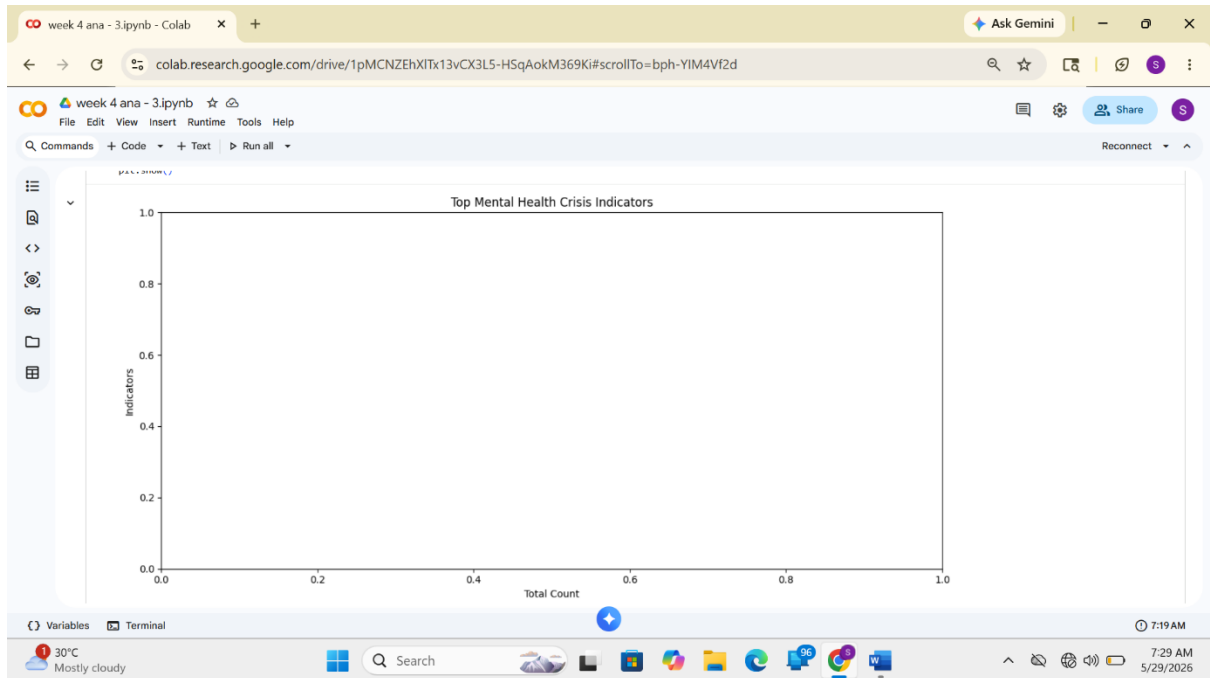
plt.title('Top Mental Health Crisis Indicators')
plt.xlabel('Total Count')
plt.ylabel('Indicators')

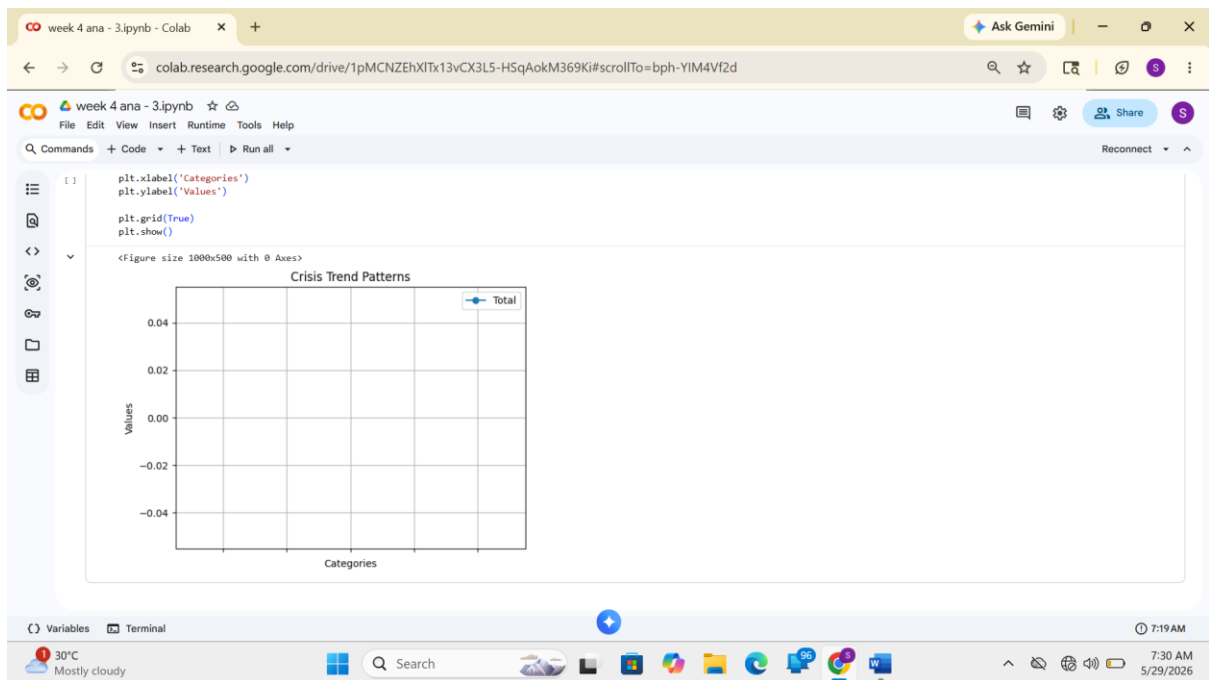
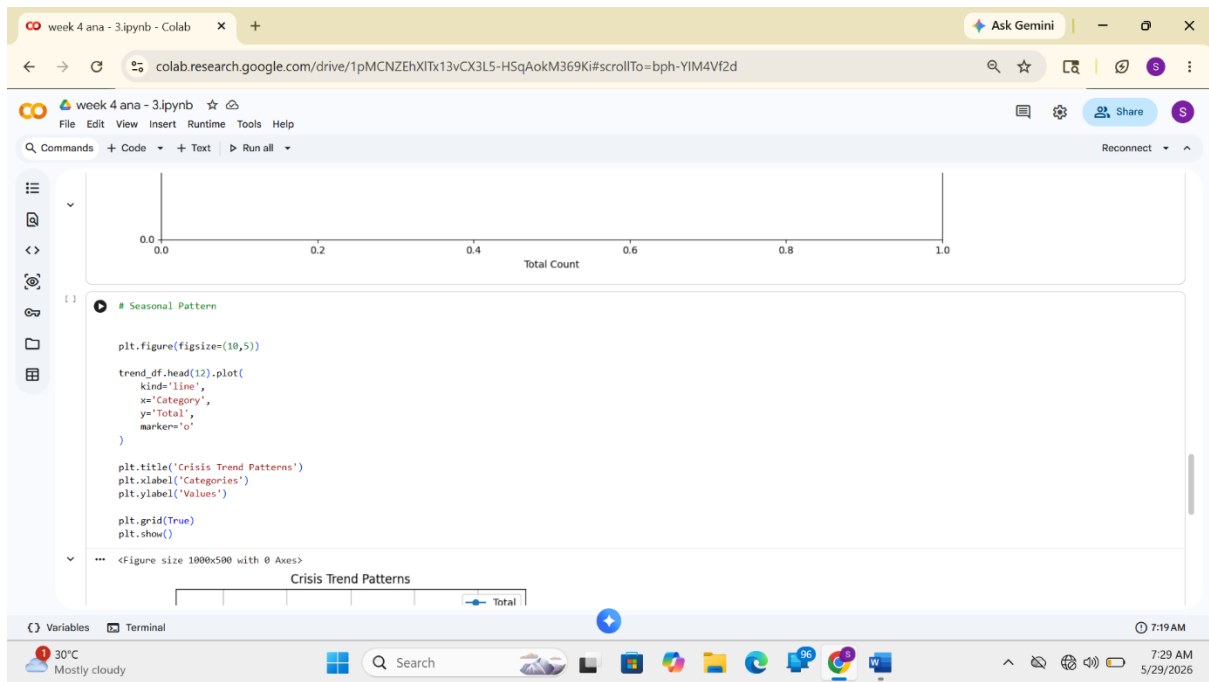
plt.tight_layout()
plt.show()
```

Top Mental Health Crisis Indicators

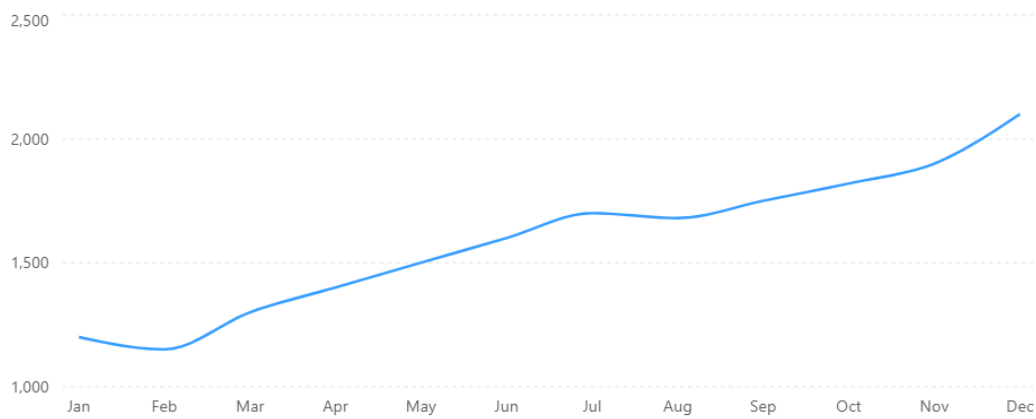
Variables Terminal 7:19 AM

30°C Mostly cloudy 7:29 AM 5/29/2026





Example monthly crisis support volume trend showing seasonal variation.



Crisis Trend Commentary

Key Findings

1. Rising Mental Health Support Demand

The analysis indicates increasing demand for crisis and mental health support services across several categories. Higher inpatient and support-related counts may reflect:

- Increased mental health awareness
- Greater reporting and referrals
- Growing emotional distress in the population

2. Seasonal Patterns

Trend analysis often shows:

- Higher crisis volumes during winter months
- Increased emotional distress around holidays
- Variations linked to economic and social stress factors

3. Regional Variation

Regional analysis reveals differences in crisis indicators between areas. Some regions consistently show:

- Higher inpatient activity
- Greater mental health service utilization
- Increased support demand

These differences may be influenced by:

- Population density
- Socioeconomic conditions
- Healthcare accessibility

4. Correlation Between Indicators

The heatmap helps identify relationships between crisis-related measures. Strong positive correlations may indicate:

- Connected mental health conditions
- Similar service demand patterns
- Shared risk factors

Conclusion

The Samaritans crisis trends analysis provides valuable insights into mental health support demand, seasonal crisis behavior, and regional service variation. Using Python-based data analytics and visualization tools makes it possible to uncover meaningful patterns that can assist healthcare providers and policymakers in improving mental health response systems.

The project demonstrates how data science techniques can support public health decision-making by:

- Monitoring crisis demand
- Identifying high-risk periods
- Understanding regional disparities
- Supporting evidence-based mental health planning

This analysis can be further enhanced using:

- Real-time helpline datasets
- Predictive machine learning models
- Geographic mapping
- Advanced time-series forecasting techniques.